

EE 412/512 Project Part 1

Due November 25, 2025

Design the assigned amplifier using a single BFU520W (a BJT). Use ADS for simulation. Refer to ADS Tutorials and related documentation.

Source and load resistances are both 50Ω .

Low-noise amplifier

Unconditionally stable at all frequencies (40MHz to 2GHz)

Bandwidth: 10% of the center frequency

Input return loss $> 8\text{dB}$

Output return loss $> 12\text{dB}$

Gain $> 16 - 6.5 \times f \text{ (GHz) dB}$ (f is the center frequency)

Gain variation $< \pm 0.5\text{dB}$ in the band

Noise figure $< 0.6 + 0.33 \times f \text{ (GHz) dB}$ (f is the center frequency)

High-gain amplifier

Unconditionally stable at all frequencies (40MHz to 2GHz)

Bandwidth: 10%

Input return loss $> 12\text{dB}$ in the band

Output return loss $> 12\text{dB}$ in the band

Gain $> 19 - 6.5 \times f \text{ (GHz) dB}$ (f is the center frequency)

Gain variation $< \pm 0.5\text{dB}$ in the band

Choose the operating point according to gain or noise considerations.

Design stabilization network(s), input and output matching networks of your circuit.

In your PDF report, provide

- A nice-looking schematic generated by ADS.
- Stability factor K at all frequencies (0.04-2GHz) showing clearly that it is greater than 1 (you may need to set the vertical axis manually).
- S11 (dB) and S22 (dB) simulations in your band showing that they satisfy the requirements. Add markers to the worst points.
- S21 (dB) in your band with three markers showing the gain variation
- Noise figure NF (dB) in the band with a marker showing the highest noise figure.

Please name your report as YourLastName.pdf.